Azure Default Outbound Access (For VMs Without Public IPs) When an Azure Virtual Machine does not have a public IP address, Azure still allows outbound internet connectivity through a temporary public IP under the following conditions: ✅ Conditions for Default Outbound Access: The VM does not have a Public IP assigned. There is no NAT Gateway configured for outbound traffic. There is no Azure Load Balancer with outbound rules applied. 🚨 Important Notes: This behavior is provided by Azure SNAT (Source Network Address Translation). The outbound IP is dynamic and can change anytime. It is not recommended for production because it's not a stable or dedicated outbound IP. Microsoft plans to remove this feature in the future, encouraging users to configure a NAT Gateway or Load Balancer outbound rules for predictable outbound IPs. Recommended Solutions for Reliable Outbound Access: 1️⃣ Assign a Public IP to the VM → If direct outbound connectivity is needed. 2️⃣ Use a NAT Gateway → Best practice for stable and scalable outbound connectivity. 3️⃣ Configure an Azure Load Balancer with outbound rules → If multiple VMs need outbound access. what is the use of in that case nat? In Azure, a NAT Gateway (Network Address Translation Gateway) is used to provide stable, scalable, and controlled outbound internet access for virtual machines (VMs) in a private subnet. Why Use a NAT Gateway Instead of Default Outbound Access? By default, Azure provides temporary outbound access for VMs without a public IP (via Default Outbound Access). However, this is unreliable because: ❌ The outbound IP address can change anytime. ❌ Microsoft may remove this default behavior in the future. ❌ It doesn’t support large-scale outbound traffic efficiently. Benefits of Using a NAT Gateway in Azure ✅ Static Outbound IP Address: You can assign a fixed public IP to outbound traffic. ✅ High Availability: NAT Gateway is fully managed by Azure and scales automatically. ✅ Better Performance: Supports 64,000 concurrent flows per IP and up to 16 public IPs for large-scale outbound traffic. ✅ Security: No inbound connections are allowed (unlike traditional public IPs on VMs). How NAT Gateway Works in Azure Outbound traffic from VMs in a private subnet goes through the NAT Gateway. The NAT Gateway replaces the VM’s private IP with a public IP before sending traffic to the internet. This ensures that all outbound connections from those VMs appear to come from a single, static public IP. When Should You Use a NAT Gateway? 🔹 If your VMs don’t have public IPs but need stable outbound internet access. 🔹 When you need multiple VMs in a private subnet to use the same fixed outbound IP. 🔹 To prevent Azure from changing outbound IPs, ensuring consistent external communication. 🔹 If you want better performance and security compared to default outbound access.